



ScholarQuest: Gamified Learning Platform

Master of Science in Information Technology (MSc IT) Project Report • Academic Year 2025–2026



1. Executive Abstract

ScholarQuest is an innovative, full-stack gamified learning and study management web application designed to transform traditional academic productivity into an immersive Role-Playing Game (RPG) experience. Students often struggle with motivation, study consistency, and burnout when preparing for technical coursework and software development challenges.

ScholarQuest addresses this problem by integrating core study utilities—such as interactive battle quizzes (**QuizForge**), RPG Pomodoro focus timers with ambient audio (**Focus Arena**), daily study habit tracking (**HabitDojo**), real-time collaborative study rooms (**StudySync**), and an interactive code compiler (**CodeScroll**)—into a unified gamified ecosystem.



2. System Modules & Functionalities

#	Module Name	Core Functionality & Features
1	Authentication & Identity Launcher	• Dual-identifier login (accepts both Email Address and Username). • Account registration with salted password hashing (bcryptjs). • 6-digit OTP password recovery workflow. • Social SSO authentication integration (Google, GitHub, LeetCode). • Guest Preview Mode & persistent JWT session management with offline fallback.
2	Hero Avatar & Progression System	• 4 Base Selectable Classes: <i>The Scholar, The Apprentice, The Builder, The Explorer</i>. • 17 Unlockable Heroes: Spanning Combat, Royalty, Gold, Habit, Focus, and StudySync tiers. • Individual solo standing character sprite cropping. • Live real-time synchronization across Top Header, Hero Showcase Banner, and Sidebar Widget.
3	QuizForge (Battle Quizzes)	• Gamified timed quiz battles across Data Structures, Algorithms, and MSc IT subjects. • Real-time score calculation, XP gains, and Gold coin rewards for correct answers.
4	Focus Arena (RPG Pomodoro Timer)	• Customizable focus timer (25m / 50m / 90m sessions). • Integrated ambient soundscapes (<i>Lofi Realm, Rain & Thunder, Tavern Hearth, Medieval Library</i>). • Focus streak multipliers and automatic XP/HP reward allocation upon session completion.
5	HabitDojo (Habit Multiplier)	• Daily study habit tracking with difficulty tiers (<i>Easy, Medium, Hard</i>). • Consecutive habit streak tracking and multiplier calculation.
6	StudySync (Collaborative Study)	• Virtual study rooms and real-time scholar chat. • Peer status indicators and active room connection metrics.
7	CodeScroll (Interactive Compiler)	• In-browser live code execution environment and syntax-highlighted editor. • Interactive programming exercises and compiler feedback.
8	Shop & Inventory System	• In-game RPG marketplace to purchase unlockable hero classes and visual themes (<i>Gold, Emerald, Cyber, Obsidian</i>). • Inventory item tracking and badge management using earned Gold coins.



3. Conclusion

ScholarQuest successfully demonstrates the integration of gamification theory with modern web engineering. By leveraging a responsive modular architecture, robust dual-identifier authentication, offline fallback resilience, and an engaging RPG progression model, the platform provides an effective, high-yield study tool tailored for MSc IT students and software learners.